

Notes: Li, Ng, and Swaminathan (JFE, 2013)

Predicting Market Returns Using Aggregate Implied Cost of Capital

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7.3 Takeaway

1 Big picture

- **Question.** Does the aggregate implied cost of capital (ICC) forecast future excess market returns, and does it do so better than the standard predictors used in the return-predictability literature?
- **Setting.** The paper constructs a monthly aggregate ICC for the U.S. equity market from January 1977 to December 2011 using firm-level ICCs for prevailing S&P 500 constituents.
- **Why this matters.** Traditional predictors such as dividend yield, earnings yield, book-to-market, and business-cycle spreads often look promising in-sample but perform weakly once one corrects for persistence, overlapping horizons, small samples, or out-of-sample evaluation.
- **Core conceptual contribution.** The paper introduces the *aggregate ICC* as a theoretically disciplined forecasting variable. Unlike a simple valuation ratio, ICC uses current prices *and* explicit expectations of future cash flows and growth.
- **Key theoretical point.** Under the Pastor, Sinha, and Swaminathan logic, if expected returns are time-varying and approximately AR(1), aggregate ICC should be an excellent proxy for the market's conditional expected return.
- **Main empirical strategy.** Estimate firm-level ICCs from analyst earnings forecasts and a finite-horizon free-cash-flow valuation model, aggregate them to the market level, and compare their forecasting power against valuation ratios and business-cycle variables in both in-sample and out-of-sample tests.
- **Main baseline result.** ICC positively predicts future excess market returns at all horizons from one month to four years. In univariate monthly regressions, the ICC slope is significant at every horizon, with adjusted R^2 rising from about 1% at one month to 31% at forty-eight months.
- **Horse-race result.** ICC remains significant when paired with D/P, E/P, B/M, P/Y, the term spread, the default spread, the short rate, and the long Treasury yield. The competing variables generally lose significance once ICC is included.
- **Portfolio-level result.** ICCs for size portfolios and book-to-market portfolios predict the corresponding portfolio returns, which shows that the forecasting power is not confined to the aggregate index alone.
- **Out-of-sample result.** Using a two-year moving average of ICC, the out-of-sample R^2_{OS} from 1998–2011 is 1.22% with a Clark–West p -value of 0.06, and the implied utility gain for a mean–variance investor is 4.15% per year.
- **Economic takeaway.** A valuation-based measure that embeds expected growth performs much better than standard raw ratios in forecasting returns. The paper's main message is that ICC contains cleaner information about time-varying expected returns.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- The return-predictability debate is an ideal environment in which to test whether a theoretically motivated expected-return proxy adds real information beyond reduced-form ratios.
- ICC is especially attractive because it comes directly from a valuation model rather than from a loose empirical regularity.
- The market-level setting is also useful because aggregation should attenuate firm-level ICC measurement error. The paper explicitly argues that if ICC works better after aggregation, then it should also work for diversified portfolios.
- The paper studies the exact predictors that dominate the literature, so the exercise is a direct comparison rather than a new predictor in an otherwise different environment.

Interpretation. The contribution is not merely “another predictor.” The paper tries to show that a more structural expected-return proxy outperforms the older valuation-ratio approach when both are held to the same statistical standard.

2.2 Data sources and sample construction

- **CRSP:** market capitalization and returns.
- **Compustat:** accounting variables such as dividends, earnings, book value, and fiscal-year timing.
- **I/B/E/S:** analyst earnings forecasts and prices.
- **BEA:** nominal GDP growth, used to anchor long-run growth assumptions.

The sample spans **January 1977 to December 2011**. For the main aggregate ICC, each month the authors compute firm-level ICCs for the firms that are in the S&P 500 at that date and then form a **value-weighted average**. To ensure public-information timing, the paper uses accounting data from the most recent fiscal year ending at least **three months** before the month-end at which ICC is computed.

To mitigate outliers, the monthly cross-section of firm-level ICCs is trimmed by removing observations more than **five standard deviations** away from the monthly mean. The paper reports that the main results are robust to not trimming.

2.3 What the ICC is measuring

The starting point is the infinite-horizon dividend discount model:

$$P_t = \sum_{k=1}^{\infty} \frac{E_t(D_{t+k})}{(1 + r_e)^k}.$$

Using the Campbell–Shiller log-linear approximation, the paper defines the ICC as the value of $r_{e,t}$ satisfying

$$r_{e,t} = \kappa + (1 - \rho)(d_t - p_t) + (1 - \rho)E_t \sum_{j=0}^{\infty} \rho^j \Delta d_{t+1+j}.$$

This expression shows the key intuition:

- a higher current dividend yield pushes the ICC up;
- stronger expected future dividend growth also pushes the ICC up;
- hence ICC is a valuation object that mixes price, payout, and growth expectations.

Interpretation. A raw ratio like D/P captures only current valuation. ICC explicitly adds information about the *expected path of future cash flows*, which is why the paper expects it to dominate traditional ratios.

2.4 Firm-level ICC construction

The empirical implementation uses a finite-horizon free-cash-flow model:

$$P_t = \sum_{k=1}^T \frac{FE_{t+k}(1 - b_{t+k})}{(1 + r_e)^k} + \frac{FE_{t+T+1}}{r_e(1 + r_e)^T},$$

where FE_{t+k} is the earnings forecast for year $t + k$, b_{t+k} is the plowback rate, and the baseline horizon is $T = 15$.

The construction proceeds in stages:

- **One-year-ahead earnings.** FE_1 is a weighted average of I/B/E/S FY1 and FY2 forecasts, where the weights depend on how many months remain until the firm's next fiscal year-end.
- **Two-year-ahead earnings.** FE_2 is built from the growth rate implicit in FY2 relative to FY1.
- **Longer-horizon earnings.** From year $t + 3$ onward, earnings growth mean-reverts exponentially toward the long-run nominal GDP growth rate by year $t + T + 2$.
- **Plowback.** The current plowback rate starts from one minus the latest payout ratio and then reverts linearly toward a steady-state plowback rate of g/r_e .
- **Terminal value.** Beyond year T , the paper uses a no-economic-profit perpetuity. That does *not* mean cash flows stop growing; it means incremental investment after the horizon earns just the cost of capital.

The paper also caps or fills extreme forecast-growth cases so that the procedure remains well defined when analyst forecast inputs are missing or implausibly large.

Interpretation. ICC is not a mechanical transformation of price. It is a model-based discount rate backed out from price after feeding in an explicit path for future earnings and payout behavior.

2.5 Aggregate ICC and alternative market-level predictors

The baseline aggregate ICC is the **value-weighted average** of firm-level ICCs for all firms in the S&P 500 in a given month, minus the one-month T-bill yield. The paper refers to this excess version simply as **ICC**.

The competing predictors are:

- valuation ratios: **D/P**, **E/P**, **B/M**, **P/Y**;

- business-cycle variables: **Term, Default, Tbill, Yield**.

Definitions follow the standard literature:

- D/P, E/P, and B/M are value-weighted averages of firm-level S&P 500 ratios;
- P/Y is payout yield from Boudoukh et al.;
- Term is the AAA yield minus the one-month T-bill rate;
- Default is the BAA minus AAA spread;
- Tbill is the one-month Treasury bill rate;
- Yield is the 30-year Treasury yield.

2.6 Key summary statistics

Table 1 is very revealing.

- Average annualized ICC is **7.07%**; its standard deviation is **2.68%**.
- Average annualized realized excess market return is **6.28%**, so the level of ICC is economically plausible.
- The first-order autocorrelation of ICC is **0.95**, but it drops to **0.10** by 24 months and becomes negative by 36 months.
- By contrast, D/P, E/P, and B/M have first-order autocorrelations between **0.98 and 0.99** and remain above roughly **0.40** even at 60 months.
- The paper reports that unit-root tests reject a unit root for ICC but not for the main valuation ratios.
- ICC is positively correlated with the valuation ratios and with Term and Default, which suggests it contains information about both valuation and the business cycle.

Interpretation. ICC is persistent, but not *too* persistent. That matters because highly persistent predictors often generate misleading long-horizon predictability and fragile inference.

3 Models and empirical specifications (all equations + interpretation)

3.1 Why ICC should proxy for expected returns

The paper relies on a theoretical result from Pastor, Sinha, and Swaminathan: if the conditional expected return follows an AR(1) process, then the aggregate ICC is perfectly correlated with the true conditional expected return.

This yields two testable implications:

- high ICC should predict high future excess returns;

- ICC should outperform simpler valuation ratios because it uses growth information those ratios omit.

Interpretation. The paper is not claiming that ICC is an unrelated statistical predictor. It is claiming that ICC is close to the object the literature actually wants: the market’s time-varying expected return.

3.2 In-sample forecasting regressions

The main in-sample forecasting specification is the long-horizon regression

$$\frac{1}{K} \sum_{k=1}^K r_{t+k} = a + bX_t + u_{t+K,t},$$

where r_{t+k} is the continuously compounded excess return and X_t is a forecasting variable such as ICC, D/P, E/P, B/M, P/Y, Term, Default, Tbill, or Yield.

The paper estimates this regression at multiple horizons:

- monthly horizons: $K = 1, 12, 24, 36, 48$ months;
- quarterly horizons: $K = 1, 4, 8, 12, 16$ quarters.

The note focuses mainly on the monthly specifications because those are the ones emphasized in the paper’s tables.

3.3 Inference and small-sample corrections

The econometric issues are standard but important:

- long-horizon regressions use **overlapping observations**;
- this induces serial correlation in residuals;
- residuals can also be conditionally heteroskedastic.

To address this, the paper computes **GMM/Newey–West** standard errors with $K - 1$ lags. It also constructs an **average slope statistic** across horizons, following Richardson and Stock, to test the null that the predictive slopes are jointly zero.

Most importantly, the paper does not rely only on asymptotic Z -statistics. It generates **Monte Carlo finite-sample p -values** from a VAR-based data-generating process under the null of no predictability. This is one of the paper’s strongest design choices.

Interpretation. The paper takes seriously the usual objections to return-predictability regressions: persistence, overlap, and small-sample bias. ICC’s forecasting power is not just an artifact of naive asymptotic inference.

3.4 Bivariate horse races

To test whether ICC merely repackages existing predictors, the paper estimates bivariate versions of the same long-horizon regressions with

$$X_t \in \left\{ \begin{array}{l} (ICC, D/P), (ICC, E/P), (ICC, B/M), \\ (ICC, P/Y), (ICC, Term), (ICC, Default), \\ (ICC, Tbill), (ICC, Yield) \end{array} \right\}.$$

The logic is simple:

- if ICC only duplicates old information, it should become insignificant once the old variables are included;
- if ICC truly contains cleaner expected-return information, it should remain significant and the other variables should not.

3.5 Portfolio-level forecasting tests

As an additional check, the paper forms:

- three **size portfolios**: Small, Medium, Large;
- three **book-to-market portfolios**: Low B/M, Medium B/M, High B/M.

For each portfolio, the authors compute portfolio-level ICCs and traditional valuation measures as value-weighted averages of firm-level quantities, and then run the same long-horizon predictive regressions on the corresponding portfolio excess returns.

Interpretation. This is an aggregation argument. If firm-level noise is the problem, then portfolio-level ICC should also work well, and the paper shows that it does.

3.6 Out-of-sample forecasting specification

For out-of-sample tests, the paper moves to the one-step-ahead predictive regression

$$r_{t+1} = \alpha_i + \beta_i x_{i,t} + \varepsilon_{i,t+1}.$$

Forecasts are generated recursively with an expanding estimation window. The benchmark forecast is the historical average excess return. Because in-sample evidence is stronger at longer horizons, the paper uses a **two-year moving average of ICC** as the out-of-sample ICC predictor.

The main out-of-sample statistic is

$$R_{OS}^2 = 1 - \frac{\sum_{k=1}^q (r_{m+k} - \hat{r}_{i,m+k})^2}{\sum_{k=1}^q (r_{m+k} - \bar{r}_{m+k})^2},$$

where $\hat{r}_{i,m+k}$ is the recursive forecast from predictor i and $\bar{r}_{m+k}$ is the historical-average forecast.

The paper evaluates out-of-sample accuracy using three tools:

- R_{OS}^2 relative to the historical average;

- the **Clark–West adjusted MSPE test** for nested forecasts;
- **economic value** for a mean–variance investor.

The portfolio allocation rule is

$$w_{i,t} = \frac{1}{\gamma} \frac{\hat{r}_{i,t+1}}{\hat{\sigma}_{t+1}^2},$$

with risk aversion $\gamma = 3$, a rolling ten-year variance estimate, and portfolio weights constrained to lie between 0 and 1.5.

3.7 Forecast-encompassing tests

To compare ICC more directly with alternative predictors, the paper uses the Harvey–Leybourne–Newbold forecast-encompassing statistic. The null is that the forecast from variable R encompasses the forecast from variable C .

Interpretation. Encompassing matters because a predictor can have a positive out-of-sample R^2 yet still add no information once another forecast is already in hand. The paper shows that ICC tends to encompass others rather than the reverse.

4 Identification and instruments (what makes the estimates credible)

4.1 What is being “identified” here

This paper is not a causal treatment-effect design. There is no instrument, no quasi-experiment, and no policy discontinuity. What the paper tries to identify is something different: whether ICC is a *credible proxy for time-varying expected returns* and whether it contains incremental predictive information relative to the standard predictors.

Interpretation. The credibility question is not “is b causal?” It is “does ICC genuinely measure conditional expected returns better than the alternatives?”

4.2 Why the paper’s evidence is persuasive

- **Theory discipline.** ICC comes from an explicit valuation model rather than an ad hoc transformation of prices.
- **Growth information.** Because ICC uses analyst forecasts and long-run growth assumptions, it can distinguish between a high expected return and a high price justified by strong expected cash-flow growth.
- **Persistence problem is milder.** ICC is still persistent, but much less persistent than D/P, E/P, and B/M, which reduces the standard Stambaugh-style concern that long-horizon predictability is being driven by near-unit-root regressors.
- **Finite-sample inference.** The Monte Carlo p -values directly target the small-sample environment of persistent predictors and overlapping returns.

- **Horse races.** ICC retains significance when the classic predictors are included jointly two at a time.
- **Out-of-sample performance.** Many return predictors look better in-sample than out-of-sample. ICC still beats the historical-average benchmark out of sample.
- **Economic significance.** The investor-utility calculations show that the forecasting gains are not merely statistically detectable; they are large enough to matter for dynamic asset allocation.

4.3 Main empirical credibility results

The strongest credibility results are:

- In **Table 2**, ICC is significant at every horizon, while D/P, E/P, B/M, and most business-cycle variables are weak or insignificant once proper finite-sample inference is used.
- In **Table 3**, ICC remains significant in every bivariate horse race, with average-slope p -values around 0.04–0.05 against valuation ratios and around 0.01–0.02 against the business-cycle variables.
- In **Table 4**, the results survive different forecast horizons ($T = 10$ and $T = 20$), inclusion of repurchases, equal-weight aggregation, all-firm aggregation, use of the long Treasury yield, and an alternative Easton-style ICC construction.
- In **Table 6**, ICC is the only variable with a clearly positive and statistically supported out-of-sample performance.

4.4 What the paper cannot fully separate

- The paper does **not** distinguish sharply between rational time-varying discount rates and time-varying market mispricing. The authors are explicit that they remain agnostic on the source of predictability.
- The paper relies on **analyst forecasts**. If those forecasts are biased or slow-moving, the measured ICC will inherit some of that error.
- The long-run growth and payout assumptions are economically sensible but still **modeling assumptions**, not directly observed objects.
- The out-of-sample evidence is strong by the standards of this literature, but it is not overwhelming in a universal sense: the headline p -value is **0.06**, so the statistical margin is respectable rather than huge.

4.5 Relation to the literature

The paper sits at the intersection of:

- the classic market-return predictability literature based on valuation ratios;
- the ICC literature that had previously focused more on unconditional equity premia, cost-of-equity estimation, and cross-sectional applications;

- the business-cycle predictability literature that emphasizes spreads and interest rates.

Its distinctive contribution is to show that a valuation-implied expected-return measure dominates the standard reduced-form predictors both statistically and economically.

5 Tables and figures

5.1 Table 1: ICC is less persistent than standard valuation ratios

Read:

- ICC has a mean of 7.07% and a standard deviation of 2.68%.
- Its first-order autocorrelation is 0.95, but it falls sharply with horizon and becomes negative by 36 months.
- D/P, E/P, and B/M are much more persistent and remain highly autocorrelated even at five-year lags.
- The average realized excess market return is 6.28%, close to the average ICC.

Economic message. This table explains why ICC is statistically attractive. It is persistent enough to forecast, but not so persistent that it behaves like an almost-unit-root ratio.

Table 1

Summary statistics for forecasting variables.

This table provides mean, standard deviation, and autocorrelations of forecasting variables used in the paper. The NYSE/Amex/Nasdaq value-weighted market excess returns (*Vwretd*), the aggregate implied cost of capital (*ICC*), the dividend-to-price ratio (*D/P*), the earnings-to-price ratio (*E/P*), the book-to-market ratio (*B/M*), the term spread (*Term*), the default spread (*Default*), the one-month T-bill rate (*Tbill*), and the 30-year Treasury yield (*Yield*) are monthly data from January 1977 to December 2011; and the payout yield (*P/Y*) (in logarithm) is monthly data from January 1977 to December 2010. All variables except *B/M* and *P/Y* are reported in annualized percentages. The autocorrelations for *Vwretd* represent the sum of autocorrelations up to the given lag, and the autocorrelations for all other variables represent the autocorrelations at the specific lag. Detailed descriptions for these variables are provided in [Section 3](#).

Variable	Mean	Standard deviation	Autocorrelation at lag					
			1	12	24	36	48	60
<i>ICC</i>	7.07	2.68	0.95	0.56	0.10	−0.20	−0.25	−0.09
<i>Vwretd</i>	6.28	15.92	0.09	0.02	−0.24	−0.23	−0.36	−0.51
<i>D/P</i>	2.61	1.05	0.98	0.87	0.79	0.73	0.64	0.52
<i>E/P</i>	6.88	2.53	0.99	0.84	0.73	0.65	0.54	0.40
<i>B/M</i>	0.45	0.18	0.99	0.88	0.82	0.74	0.65	0.53
<i>P/Y</i>	−2.26	0.25	0.98	0.71	0.57	0.39	0.37	0.29
<i>Term</i>	3.06	1.59	0.91	0.43	0.08	−0.27	−0.36	−0.14
<i>Default</i>	1.11	0.48	0.96	0.47	0.29	0.19	0.08	0.08
<i>Tbill</i>	5.23	3.34	0.97	0.78	0.56	0.38	0.28	0.31
<i>Yield</i>	7.24	2.60	0.99	0.87	0.77	0.69	0.56	0.45

Figure 1: Table 1 from the paper: summary statistics for the forecasting variables.

5.2 Figure 1: ICC rises in bad times and spikes during the crisis

Read:

- ICC is elevated in recessions and exceptionally high during the 2008–2009 crisis.
- It reaches about 12.8% in March 2009.

- It remains above its historical median even in late 2011.

Economic message. The figure gives the macro-finance intuition behind the forecasting regressions: when required returns are high, market prices are low relative to expected cash flows, and future returns should be higher.

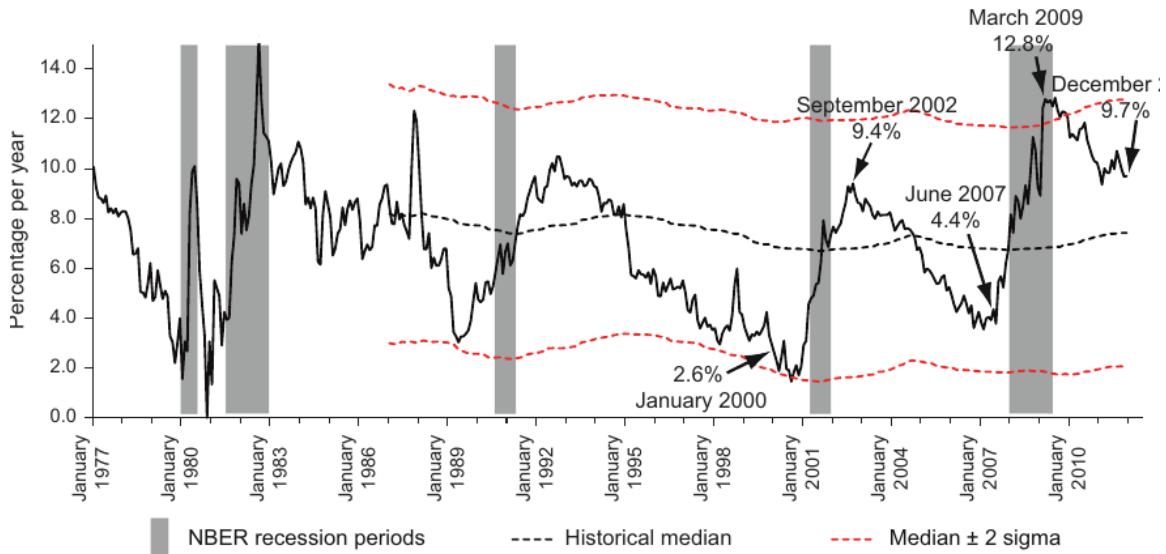


Figure 2: Figure 1 from the paper: aggregate ICC over time.

5.3 Table 2: univariate forecasting regressions

Read:

- ICC is significant at every horizon from 1 to 48 months.
- The ICC adjusted R^2 rises from 0.01 at one month to 0.31 at forty-eight months.
- Among valuation ratios, only payout yield has some predictive power at longer horizons, and even there the joint evidence is weaker than for ICC.
- Among business-cycle variables, the term spread is the only strong univariate competitor, mainly at long horizons.

Economic message. ICC is the only variable that looks consistently strong once the paper uses finite-sample Monte Carlo p -values rather than relying mechanically on asymptotic t -statistics.

Table 2

Univariate regressions for the ICC, valuation ratios, and business cycle variables.

This table summarizes univariate forecasting regression results in Eq. (6). The dependent variable in these regressions is average monthly market excess returns, and the independent variables are the aggregate implied cost of capital (*ICC*), the dividend-to-price ratio (*D/P*), the earnings-to-price ratio (*E/P*), the book-to-market ratio (*B/M*), the payout yield (*P/Y*), the term spread (*Term*), the default spread (*Default*), the one-month T-bill rate (*Tbill*), and the 30-year Treasury yield (*Yield*), in Panels A–I, respectively. *P/Y* is in logarithm form and spans January 1977 to December 2010. All other variables span January 1977 to December 2011. *K* is the forecasting horizon in months. In forecasting horizons beyond one month, the regressions use overlapping observations. *b* is the slope coefficient from the ordinary least squares (OLS) regressions. *Average* is the average slope coefficient. *Z(b)* is the asymptotic *Z*-statistics computed using the generalized method of moment (GMM) standard errors with *K*–1 Newey–West lag correction. The *adj.R*² is obtained from the OLS regression. The *p*-values (*pval*) of *Z*-statistics and the average slope coefficient are obtained by comparing the test statistics with their empirical distribution generated under the null of no predictability from five thousand trials of a Monte Carlo simulation as described in the Appendix.

<i>K</i>	<i>b</i>	<i>Z(b)</i>	<i>pval</i>	<i>adj.R</i> ²
Panel A: <i>ICC</i>				
1	2.02	1.85	0.052	0.01
12	1.63	2.30	0.076	0.07
24	1.77	2.52	0.079	0.17
36	1.77	3.18	0.048	0.27
48	1.55	3.90	0.035	0.31
Average	1.75		0.020	
Panel B: <i>D/P</i>				
1	3.42	1.30	0.221	0.00
12	3.42	1.59	0.271	0.05
24	2.74	1.38	0.352	0.07
36	2.25	1.42	0.379	0.08
48	2.24	1.88	0.316	0.13
Average	2.81		0.349	
Panel C: <i>E/P</i>				
1	1.20	1.06	0.320	0.00
12	1.13	1.28	0.359	0.03
24	0.89	1.12	0.440	0.04
36	0.71	1.05	0.491	0.05
48	0.66	1.23	0.481	0.07
Average	0.92		0.441	
Panel D: <i>B/M</i>				
1	0.16	1.04	0.365	0.00
12	0.17	1.40	0.365	0.03
24	0.13	1.17	0.470	0.05
36	0.11	1.18	0.499	0.06
48	0.10	1.37	0.485	0.08
Average	0.13		0.519	
Panel E: <i>P/Y</i>				
1	0.79	0.75	0.245	0.00
12	1.13	1.42	0.161	0.04
24	1.18	1.85	0.119	0.11
36	0.96	2.33	0.089	0.13
48	0.86	2.99	0.064	0.16
Average	0.98		0.156	
Panel F: <i>Term</i>				
1	0.96	0.53	0.312	0.00
12	1.79	1.53	0.129	0.03
24	2.17	2.36	0.056	0.10
36	2.07	3.17	0.019	0.14
48	1.64	3.01	0.037	0.14
Average	1.73		0.067	
Panel G: <i>Default</i>				
1	–2.29	–0.29	0.683	0.00
12	3.59	0.80	0.357	0.01
24	2.58	0.73	0.385	0.01
36	1.67	0.54	0.453	0.01
48	2.41	0.72	0.428	0.02
Average	1.59		0.470	

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Table 2 (continued)

<i>K</i>	<i>b</i>	<i>Z(b)</i>	<i>pval</i>	<i>adj.R</i> ²
Panel H: <i>Tbill</i>				
1	–0.39	–0.46	0.308	0.00
12	–0.28	–0.37	0.362	0.00
24	–0.24	–0.70	0.288	0.00
36	–0.07	–0.20	0.429	0.00
48	0.13	0.39	0.577	0.00
Average	–0.17		0.380	
Panel I: <i>Yield</i>				
1	–0.25	–0.22	0.589	0.00
12	0.40	0.38	0.385	0.00
24	0.64	0.92	0.262	0.02
36	0.82	1.46	0.180	0.06
48	0.89	1.98	0.132	0.10
Average	0.50		0.354	

Figure 3: Table 2 from the paper: univariate long-horizon forecasting results.

5.4 Table 3: ICC survives horse races

Read:

- When ICC is paired with D/P, E/P, B/M, or P/Y, ICC remains significant while the other variable usually does not.
- When ICC is paired with the term spread, default spread, T-bill rate, or long yield, ICC again remains significant.
- The average ICC slope remains economically large, typically around 1.5 to 1.9, and the joint p -values remain small.

Economic message. ICC is not just a relabeled dividend yield or spread. It appears to subsume the predictive information in the classic variables rather than being subsumed by them.

Table 3
Bivariate regressions involving the ICC, valuation ratios, and business cycle variables.
This table summarizes bivariate forecasting regression results involving the aggregate implied cost of capital (ICC) and a control variable in Eq. (5). The control variables are the dividend-to-price ratio (D/P), the earnings-to-price ratio (E/P), the book-to-market ratio (B/M), the payout yield (P/Y), the term spread (Term), the default spread (Default), the one-month T-bill rate (Tbill), and the 30-year Treasury yield (Yld30), in Panels A–H, respectively. The dependent variable in these regressions is average monthly market excess returns. P/Y is in logarithm form and spans January 1977 to December 2010. All other variables span January 1977 to December 2011. K is the forecasting horizon in months. In forecasting horizons beyond one month, the regressions use overlapping observations. b and c are the slope coefficients from the ordinary least squares (OLS) regressions. Average is the average slope coefficient. $Z(b)$ and $Z(c)$ are the asymptotic Z -statistics computed using the generalized method of moment (GMM) standard errors with $K-1$ Newey-West lag correction. The $adjR^2$ is obtained from the OLS regression. The p -values ($psol$) of Z -statistics and the average slope coefficient are obtained by comparing the test statistics with their empirical distribution generated under the null of no predictability from five thousand trials of a Monte Carlo simulation as described in the Appendix.

Panel A: Bivariate regression involving ICC and D/P							
K	ICC			D/P			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	1.79	1.58	0.100	2.24	0.82	0.325	0.01
12	1.35	1.88	0.135	2.45	1.18	0.304	0.08
24	1.57	2.83	0.058	1.52	1.08	0.337	0.19
36	1.62	3.50	0.041	1.01	1.02	0.393	0.28
48	1.36	4.00	0.032	1.16	1.51	0.312	0.33
Average	1.54		0.045	1.68		0.406	
Panel B: Bivariate regression involving ICC and E/P							
K	ICC			E/P			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	1.87	1.70	0.085	0.79	0.70	0.383	0.01
12	1.46	2.08	0.112	0.81	1.01	0.369	0.08
24	1.66	2.75	0.071	0.50	0.95	0.400	0.18
36	1.69	3.50	0.044	0.35	0.89	0.431	0.27
48	1.46	4.17	0.034	0.36	1.17	0.391	0.32
Average	1.63		0.043	0.56		0.478	
Panel C: Bivariate regression involving ICC and B/M							
K	ICC			B/M			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	1.87	1.66	0.089	0.08	0.53	0.570	0.01
12	1.41	1.98	0.136	0.11	0.94	0.424	0.07
24	1.64	2.87	0.060	0.06	0.76	0.489	0.18
36	1.67	3.58	0.038	0.04	0.65	0.532	0.27
48	1.44	3.91	0.035	0.04	0.79	0.517	0.31
Average	1.61		0.048	0.07		0.563	
Panel D: Bivariate regression involving ICC and P/Y							
K	ICC			P/Y			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	2.36	1.88	0.053	0.06	0.05	0.513	0.01
12	1.22	1.82	0.139	0.68	0.85	0.297	0.07
24	1.26	2.88	0.130	0.73	1.23	0.238	0.17
36	1.59	3.36	0.057	0.41	1.29	0.243	0.29
48	1.28	3.45	0.056	0.41	1.77	0.186	0.32
Average	1.50		0.051	0.46		0.344	
Panel E: Bivariate regression involving ICC and Term							
K	ICC			Term			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	4.44	2.51	0.009	−5.06	−1.78	0.051	0.01
12	2.15	1.66	0.158	−1.30	−0.53	0.632	0.07
24	1.96	1.60	0.197	−0.40	0.25	0.542	0.17
36	1.93	2.05	0.141	−0.31	−0.27	0.544	0.26
Average							
Panel F: Bivariate regression involving ICC and Default							
K	ICC			Default			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	2.54	2.37	0.013	−7.62	−0.07	0.837	0.01
12	1.62	2.18	0.086	0.13	0.03	0.494	0.06
24	1.88	2.81	0.053	−1.45	−0.54	0.643	0.17
36	1.92	3.42	0.038	−2.06	−0.83	0.707	0.27
48	1.58	3.58	0.040	−0.64	−0.23	0.561	0.30
Average	1.91		0.030	−2.33		0.608	
Panel G: Bivariate regression involving ICC and Tbill							
K	ICC			Tbill			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	2.14	1.90	0.043	0.24	0.28	0.646	0.00
12	1.71	2.19	0.076	0.20	0.27	0.623	0.06
24	1.84	2.32	0.085	0.20	0.26	0.699	0.17
36	1.83	3.08	0.044	0.23	0.87	0.758	0.27
48	1.62	4.00	0.020	0.35	1.50	0.856	0.32
Average	1.83		0.030	0.24		0.698	
Panel H: Bivariate regression involving ICC and Yield							
K	ICC			Yield			$adjR^2$
	b	$Z(b)$	$psol$	c	$Z(c)$	$psol$	
1	2.02	1.85	0.052	−0.33	−0.23	0.482	0.00
12	1.62	2.25	0.082	0.34	0.37	0.312	0.06
24	1.73	2.56	0.078	0.47	0.91	0.201	0.18
36	1.67	3.22	0.056	0.40	1.36	0.145	0.28
48	1.41	3.89	0.039	0.53	1.75	0.118	0.34
Average	1.69		0.021	0.32		0.362	

Figure 4: Table 3 from the paper: bivariate regressions involving ICC and other predictors.

5.5 Table 4: robustness across alternative ICC constructions

Read:

- ICC built with $T = 10$ or $T = 20$ still predicts returns.
- Incorporating repurchases does not undo the result.
- Equal-weighted ICC, all-firm ICC, yield-adjusted ICC, and Easton-style ICC all remain predictive.

Economic message. The forecasting success is not a knife-edge artifact of one exact implementation choice.

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Table 4

Alternative ICC specifications.

This table provides univariate regression of Eq. (6) for alternative measures of the aggregate implied cost of capital (ICC), ICC_{T10} , ICC_{T20} , and $ICC_{repurchase}$ are obtained with forecasting horizon $T = 10$, $T = 20$, and net share repurchases, respectively. ICC_{all} is the value-weighted average of the firm-level ICC for all firms in our sample universe subtracting the one-month T-bill yield, ICC_{equal} is the equal-weighted average of the firm-level ICC for firms in the Standard & Poor's (S&P) 500 index subtracting the one-month T-bill yield, and ICC_{yield} is the value-weighted average of the firm-level ICC for firms in the S&P 500 index subtracting the 30-year Treasury yield. This table also provides the regression results for ICC in which ICC is calculated based on the method of Easton (2004). The dependent variable in these regressions is average monthly market excess returns. All regressions use data from January 1977 to December 2011. K is the forecasting horizon in months. In forecasting horizons beyond one month, the regressions use overlapping observations. b is the slope coefficient from the ordinary least squares (OLS) regressions. Average is the average slope coefficient. $Z(b)$ is the asymptotic Z-statistics computed using the generalized method of moment (GMM) standard errors with $K-1$ Newey-West lag correction. The $adj.R^2$ is obtained from the OLS regression. The p -values ($pval$) of Z-statistics and the average slope coefficient are obtained by comparing the test statistics with their empirical distribution generated under the null of no predictability from five thousand trials of a Monte Carlo simulation as described in the Appendix.

K	b	$Z(b)$	$pval$	$adj.R^2$
Panel A: ICC_{T10}				
1	2.40	1.98	0.039	0.01
12	1.93	2.39	0.065	0.08
24	1.94	2.33	0.095	0.17
36	1.88	2.98	0.057	0.27
48	1.63	3.97	0.030	0.31
Average	1.96		0.012	
Panel B: ICC_{T20}				
1	1.85	1.79	0.056	0.01
12	1.44	2.12	0.092	0.06
24	1.62	2.43	0.084	0.15
36	1.62	2.97	0.058	0.24
48	1.42	3.62	0.042	0.27
Average	1.59		0.027	
Panel C: $ICC_{repurchase}$				
1	2.07	1.74	0.058	0.01
12	1.73	2.29	0.070	0.06
24	1.94	2.65	0.062	0.17
36	1.92	3.21	0.041	0.26
48	1.61	3.46	0.047	0.27
Average	1.85		0.017	
Panel D: ICC_{all}				
1	2.22	1.91	0.047	0.01
12	1.80	2.48	0.061	0.07
24	1.92	2.71	0.064	0.18
36	1.89	3.28	0.041	0.27
48	1.61	3.71	0.041	0.29
Average	1.89		0.015	
Panel E: ICC_{equal}				
1	1.87	1.76	0.058	0.01
12	1.32	1.86	0.121	0.04
24	1.48	2.21	0.099	0.12
36	1.50	2.98	0.056	0.20
48	1.31	3.73	0.034	0.24
Average	1.50		0.038	
Panel F: ICC_{yield}				
1	3.28	1.78	0.061	0.01
12	2.09	1.61	0.162	0.04
24	2.15	1.69	0.183	0.09
36	2.16	2.04	0.150	0.14
48	2.18	3.25	0.053	0.22
Average	2.37		0.033	
Panel G: ICC based on Easton (2004)				
1	2.07	1.74	0.058	0.01
12	1.73	2.29	0.070	0.06
24	1.94	2.65	0.062	0.17
36	1.92	3.21	0.041	0.26
48	1.61	3.46	0.047	0.27
Average	1.85		0.017	

Figure 5: Table 4 from the paper: alternative ICC specifications.

5.6 Table 5: portfolio-level ICCs also work

Read:

- ICC predicts returns for small, medium, and large size portfolios.
- ICC also predicts returns for low, medium, and high book-to-market portfolios.
- The average ICC slope ranges roughly from 1.06 to 2.10, with Monte Carlo p -values between 0.014 and 0.038.
- Traditional valuation ratios have much weaker and less systematic predictive power at the portfolio level.

Economic message. Once some firm-level noise is diversified away, ICC keeps working. That strongly supports the interpretation that the market-level ICC captures expected returns rather than statistical accident.

Table 5

Univariate regressions for size and book-to-market portfolios.

This table summarizes univariate forecasting regression results for size and book-to-market portfolios. Small Size, Medium Size, and Large Size are the three size portfolios, and Low B/M, Medium B/M, and High B/M are the three book-to-market portfolios. The dependent variable in these regressions is average monthly excess returns for the corresponding portfolio, and the independent variables are the implied cost of capital (ICC), the dividend-to-price ratio (D/P), the earnings-to-price ratio (E/P), and the book-to-market ratio (B/M) for the corresponding portfolio. All variables span January 1977 to December 2011. In forecasting horizons beyond one month, the regressions use overlapping observations. K is the forecasting horizon in months. b is the slope coefficient from the ordinary least squares (OLS) regressions. Average is the average slope coefficient. $Z(b)$ is the asymptotic Z -statistics computed using the generalized method of moment (GMM) standard errors with $K-1$ Newey-West lag correction. The $adj.R^2$ is obtained from the OLS regression. The p -values ($pval$) of Z -statistics and the average slope coefficient are obtained by comparing the test statistics with their empirical distribution generated under the null of no predictability from five thousand trials of a Monte Carlo simulation as described in the Appendix.

K	ICC				D/P				E/P				B/M			
	b	$Z(b)$	$pval$	$adj.R^2$	b	$Z(b)$	$pval$	$adj.R^2$	b	$Z(b)$	$pval$	$adj.R^2$	b	$Z(b)$	$pval$	$adj.R^2$
Panel A: Univariate regressions for Small Size portfolio																
1	4.11	2.70	0.005	0.02	4.75	0.99	0.248	0.00	2.38	1.76	0.073	0.01	0.33	1.68	0.091	0.01
12	2.68	3.21	0.017	0.11	6.49	1.94	0.129	0.06	2.62	2.87	0.053	0.12	0.44	3.53	0.016	0.16
24	1.71	1.99	0.127	0.12	4.09	2.18	0.120	0.07	1.60	2.34	0.114	0.13	0.27	2.85	0.072	0.18
36	1.28	1.68	0.195	0.12	3.01	1.94	0.190	0.07	1.30	2.57	0.126	0.15	0.21	2.75	0.103	0.19
48	0.61	1.06	0.344	0.04	2.36	1.63	0.270	0.07	1.13	2.61	0.153	0.18	0.18	2.72	0.135	0.22
Average	2.08		0.014		4.14		0.244		1.81		0.121		0.29		0.064	
Panel B: Univariate regressions for Medium Size portfolio																
1	3.80	2.64	0.009	0.02	2.17	0.58	0.416	0.00	1.95	1.44	0.167	0.01	0.33	1.59	0.122	0.01
12	2.50	3.44	0.015	0.11	4.14	1.65	0.190	0.04	1.71	2.08	0.165	0.06	0.36	3.03	0.041	0.11
24	1.92	3.01	0.041	0.21	2.40	1.35	0.305	0.05	0.95	1.65	0.284	0.06	0.21	2.27	0.152	0.11
36	1.43	3.01	0.063	0.23	1.90	1.45	0.334	0.06	0.73	1.86	0.273	0.07	0.15	2.42	0.166	0.13
48	0.82	2.73	0.094	0.13	1.55	1.73	0.315	0.07	0.59	1.96	0.291	0.08	0.11	2.39	0.189	0.12
Average	2.10		0.015		2.43		0.402		1.19		0.345		0.23		0.194	
Panel C: Univariate regressions for Large Size portfolio																
1	1.77	1.67	0.073	0.01	2.87	1.13	0.280	0.00	0.94	0.85	0.352	0.00	0.13	0.83	0.403	0.00
12	1.42	2.07	0.089	0.05	2.74	1.35	0.326	0.03	0.84	0.98	0.408	0.02	0.13	1.10	0.422	0.02
24	1.59	2.37	0.095	0.14	2.21	1.15	0.423	0.05	0.64	0.83	0.485	0.02	0.10	0.87	0.515	0.03
36	1.70	3.08	0.060	0.22	1.89	1.16	0.472	0.06	0.52	0.75	0.525	0.03	0.09	0.84	0.548	0.03
48	1.54	3.85	0.035	0.25	1.95	1.47	0.444	0.09	0.51	0.85	0.539	0.04	0.08	0.93	0.559	0.05
Average	1.60		0.025		2.33		0.418		0.69		0.424		0.11		0.497	
Panel D: Univariate regressions for Low B/M portfolio																
1	2.51	1.99	0.041	0.01	2.97	0.76	0.358	0.00	1.47	0.93	0.328	0.00	0.15	0.61	0.460	0.00
12	1.71	1.96	0.115	0.05	3.05	1.02	0.354	0.02	1.36	1.13	0.367	0.02	0.17	0.84	0.453	0.01
24	1.66	2.03	0.123	0.11	1.77	0.62	0.480	0.02	0.90	0.79	0.473	0.02	0.09	0.48	0.589	0.01
36	1.77	2.57	0.083	0.17	1.20	0.49	0.537	0.01	0.65	0.64	0.556	0.02	0.08	0.46	0.617	0.01
48	1.59	2.62	0.092	0.18	1.16	0.51	0.547	0.02	0.56	0.62	0.592	0.02	0.06	0.41	0.634	0.01
Average	1.85		0.029		2.03		0.452		0.99		0.460		0.11		0.579	
Panel E: Univariate regressions for Medium B/M portfolio																
1	1.40	1.45	0.108	0.01	1.20	0.49	0.455	0.00	0.44	0.43	0.551	0.00	0.09	0.59	0.459	0.00
12	1.20	1.85	0.119	0.04	1.74	0.96	0.391	0.01	0.50	0.68	0.527	0.01	0.15	1.43	0.274	0.03
24	1.51	3.35	0.023	0.16	1.27	0.86	0.443	0.02	0.18	0.35	0.660	0.00	0.10	1.14	0.397	0.03
36	1.51	4.01	0.021	0.28	0.97	0.78	0.488	0.02	-0.01	-0.03	0.774	0.00	0.07	0.85	0.515	0.03
48	1.16	3.82	0.036	0.25	1.14	1.05	0.458	0.05	0.02	0.04	0.757	0.00	0.05	0.68	0.590	0.02
Average	1.36		0.038		1.27		0.501		0.22		0.671		0.09		0.474	
Panel F: Univariate regressions for High B/M portfolio																
1	0.95	1.17	0.168	0.00	2.35	1.33	0.175	0.00	0.62	0.68	0.347	0.00	0.07	0.62	0.407	0.00
12	1.30	2.93	0.020	0.08	2.72	1.67	0.186	0.06	0.67	1.03	0.299	0.02	0.13	1.56	0.231	0.04
24	1.22	3.13	0.022	0.18	2.14	1.68	0.240	0.09	0.67	1.37	0.252	0.04	0.12	1.72	0.245	0.09
36	0.99	3.24	0.032	0.21	1.98	1.97	0.223	0.15	0.72	1.85	0.184	0.09	0.10	1.80	0.243	0.12
48	0.87	2.98	0.054	0.24	1.88	2.43	0.189	0.21	0.70	2.11	0.171	0.14	0.10	1.82	0.284	0.18
Average	1.06		0.023		2.21		0.215		0.68		0.221		0.10		0.248	

Figure 6: Table 5 from the paper: forecasting regressions for size and book-to-market portfolios.

5.7 Figure 2 and Table 6: out-of-sample performance

Read:

- In Figure 2, the ICC curve spends most of the out-of-sample period above zero, meaning that ICC beats the historical-average benchmark in cumulative squared prediction error.
- In Table 6, ICC achieves an out-of-sample R_{OS}^2 of 1.22% with a p -value of 0.06.
- The implied utility gain is 4.15% per year, much larger than that of D/P, E/P, or B/M.
- Forecast-encompassing tests show that ICC tends to encompass the other forecasting variables, while the reverse is rejected for most of them.

Economic message. The paper passes the hardest test in this literature. ICC does not just fit the past in-sample; it improves genuine forecast performance relative to the historical-average benchmark.

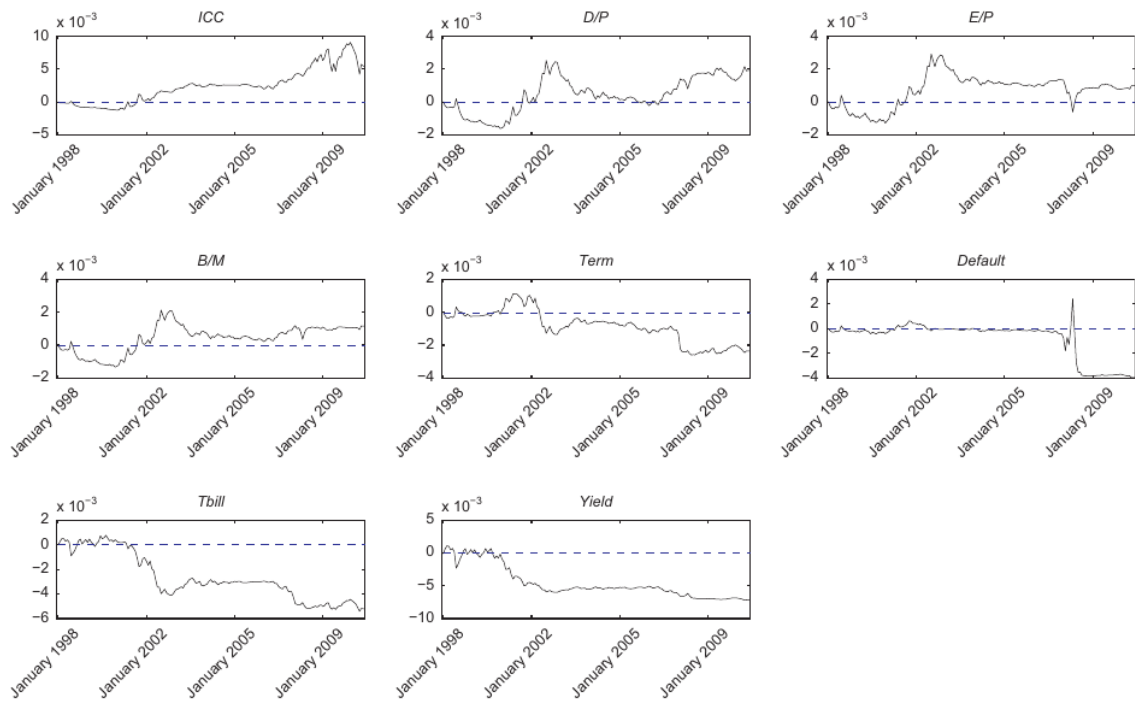


Figure 7: Figure 2 from the paper: cumulative squared prediction error relative to the historical-average benchmark.

Table 6

Out-of-sample test.

This table summarizes the out-of-sample analysis of forecasting models using different forecasting variables for the forecast period of January 1998 to December 2011. Panel A reports the R_{os}^2 statistic of Campbell and Thompson (2008). Panel A also reports the utility gain (Ugain), which is the portfolio management fee (in annualized percentage return) that an investor with mean–variance preferences and risk aversion coefficient of three would be willing to pay to have access to the forecasting model using a particular forecasting variable relative to the historical average benchmark forecasting model. The weight on stocks in the investor's portfolio is constrained to lie between zero and 1.5 (inclusive). Panel B reports the p -values of the forecasting encompassing test statistic of Harvey, Leybourne and Newbold (1998) (HLN statistic), which corresponds to a one-sided (upper-tail) test of the null hypothesis that the forecast from the row variable (R) encompasses the forecast from the column variable (C) against the alternative hypothesis that the forecast from the row variable (R) does not encompass the forecast from the column variable (C). The dependent variable in these regressions is monthly market excess returns. In these tests, we use a two-year moving average for the implied cost of capital (ICC).

Panel A: Out-of-sample R_{os}^2 and utility gains			
Variable	R_{os}^2	$pval$	Ugain
ICC	1.22	0.06	4.15
D/P	0.44	0.16	1.38
E/P	0.22	0.27	0.40
B/M	0.26	0.23	0.57
Term	−0.53		−1.32
Default	−0.90		−1.33
Tbill	−1.18		−2.21
Yield	−1.62		−1.84

Panel B: Forecast encompassing test								
Row variable (R)	Column variable (C)							
	ICC	D/P	E/P	B/M	Term	Default	Tbill	Yield
ICC		0.40	0.50	0.51	0.76	0.71	0.84	0.79
D/P	0.09		0.70	0.70	0.71	0.74	0.73	0.76
E/P	0.07	0.23		0.41	0.69	0.69	0.73	0.73
B/M	0.07	0.22	0.55		0.73	0.73	0.77	0.77
Term	0.03	0.07	0.10	0.09		0.53	0.74	0.66
Default	0.04	0.11	0.16	0.14	0.29		0.44	0.51
Tbill	0.01	0.03	0.04	0.03	0.10	0.28		0.67
Yield	0.01	0.02	0.04	0.03	0.09	0.18	0.19	

forecasting variable in the asset allocation literature, which has traditionally relied on the dividend-to-price ratio as the key forecasting variable. We remain agnostic,

hypothesis of no predictability on r_t is imposed in the VAR by setting the slope coefficients of the explanatory variables to zero and the intercept in the equation of r_t to be

Figure 8: Table 6 from the paper: out-of-sample forecasting performance and forecast-encompassing tests.

6 Short summary

- The paper asks whether the aggregate implied cost of capital is a useful forecasting variable for future excess market returns.
- It constructs firm-level ICCs from analyst forecasts, payout assumptions, and a finite-horizon valuation model, and then aggregates them to the market level.
- ICC is far less persistent than traditional valuation ratios and therefore less exposed to the usual near-unit-root problems in long-horizon forecasting regressions.
- In univariate in-sample tests, ICC predicts future returns strongly from one month to four years and dominates D/P, E/P, B/M, and most business-cycle variables.
- In bivariate tests, ICC retains predictive power while the competing predictors typically lose it.
- Alternative ICC constructions deliver the same qualitative result.
- ICCs for size and book-to-market portfolios also predict the corresponding portfolio returns.

- Out of sample, ICC is the only predictor that clearly beats the historical-average benchmark and generates economically meaningful utility gains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that aggregate ICC is a powerful predictor of future excess market returns and that it outperforms the traditional predictors that dominate the market-predictability literature.

7.2 Economic intuition

ICC works because it starts from the correct valuation identity and then backs out the discount rate implied by current prices and expected future cash flows. A high ICC means that, given the market's expected cash-flow path, investors require a high return to hold equities. That should translate into high subsequent realized returns on average.

7.3 Takeaway

If the goal is to measure time-varying expected returns, raw valuation ratios are too crude. A model-based measure that integrates prices with expected earnings growth performs much better. The big lesson of the paper is that the best return predictor in this setting is the one that is closest to the object theory actually says should matter.